

Technical Session on Correction and Exception manual and DUDBC Design Catalogue Volume 2
DUDBC Conference Hall, 6th June 2017

Participants: MoUD CL-PIU, JICA, LWF, HELVETAS, TGH, Habitat for Humanity, NOMAD, Build Change, Solve Nepal, UMN, Embassy of India, American Red Cross, Caritas, World Renew, Samaritan's Purse, AEWB (PERP-DFID), RADO- Nepal, Lumanti, HRRP.

The SDE Parikshit Kadariya, Deputy Project Director, MoUD-CLPIU chaired the session. Er. Akshaya Paudel, Engineer, CLPIU opened the meeting and introduced the agenda of the technical session on Correction and Exception manual and DUDBC Design Catalogue Volume 2.

Dr. Narayan Prasad Marasini, National Technical Coordinator, HRRP [presented](#) the findings from the technical workshop organized by HRRP district team on the all 14 earthquake affected district in the February 2017.

- Some of the major issues, challenges faced during the reconstruction process were raised by the POs during the technical workshop.
- Recommendation presented by all the participants during the workshop was discussed.
- HRRP urged all the POs to focus on the coverage of the 6 technical assistance activities. With the joint technical team consisting of technical personnel from POs, Government and HRRP the technical issues in the districts could be solved to higher extend.
- Most of the cases of non-compliance that were discussed in the technical workshop are considered to be resolve with the publication of the correction and exceptional manual by MoUD- CLPIU.

Er. Bipin Kumar Gautam, Structural Engineer, MoUD-CLPIU [presented](#) on the 11 designs published by MoUD-CLPIU for RCC design type building non-compliance with NBC 205:1994 and the Dry Stone Masonry Building.

- The detail on the design load calculation, soil type, design parameter, method of analysis considered in the designs.
- For the dry stone masonry building two types of designs was discussed i.e. Use of GI wire mess and Dressed Dry Stone Masonry with Wooden Elements.
- Information on the 3 ongoing designs by MoUD-CLPIU was given.

Er. Dipendra Aryal, Civil Engineer, MoUD-CLPIU [presented](#) on the Design Catalogue Volume 2 published by MoUD-CLPIU.

- The design published in the catalogue is for the alternative construction material and technologies.
- For the construction all the dimensions must be exactly same as indicated in the catalogue and variation in the dimensions requires different designs.
- The catalogue include 12 different technology with the information on the 3D view of the design, Floor plans, Elevation section, Structural details, Quantity estimate of major materials.

Er. Kuber Bogati, National Technical Coordination Officer, HRRP [presented](#) the correction and exception manual was published by NRA for masonry structure for houses build under the housing reconstruction programme.

- This manual is applicable only for the residential houses that have been constructed after Gorkha earthquake 2015 under the government reconstruction programme and applicable for the tranches.
- At the moment only English version has been released but soon the Nepali version will also be released.
- The need for the correction and exception manual was realized after the cases of non-compliance were observed in the field.

- This manual helps site engineers in providing advice and guidance to households for correction and to be eligible for the remaining tranches. At present only the correction and exception manual is released for the load bearing masonry structure.
- Brief information on the updated inspection form was provided along with the new form for the confined masonry building was provided.

Questions and Discussion

Q: In case of the design for differential elevation, have you considered the earth pressure especially for the houses along the roadside?

Regarding the effect of earth pressure, a retention wall between the slope and the wall of the house is constructed to resist the horizontal pressure.

Q: What is the cost comparison between CSEB and other structures?

A quantitative estimation was considered in the design so the cost may vary on the basis of the material cost in different areas.

Q: What is the rationale of having the design catalogue?

Many houses that are not approved for NBC 205:1994, so the alternative material and technology has been released.

Q: Are we considering the cultural and traditional architecture in the ongoing designs in CLPIU especially in the remote areas like Jumla, Humla?

The ongoing designs are prepared by the POs and not CLPIU itself so at the moment we are not able to say if the cultural and traditional architecture have been considered in the designs.

Q: When will the form reach district for the 2nd and the 3rd tranche?

All the forms have already reached the district except form no. 15. Please contact with the DLPIU of the respective district. The forms have been sent in abundant number to all the districts.

The SDE Parikshit Kadariya, Deputy Project Director, MoUD-CLPIU presented his closing remarks and informed that if any POs are interested in conducting trainings such as 7 days mason training or 50 days OJT, the approval letter will be issued within 15 minutes if all the necessary documents are submitted and no cases of duplication is found. He also requested everyone to constantly check the CLPIU website for regular updates on the designs. Also, in case if the new updated forms are not available in the district, the old forms can also be used and the details need to be indicated in the remarks column of the form.

Er. Akshaya Paudel closed the technical session by thanking all participants for the lively and interactive session.

Next Steps

- Training on Correction and Exception manual will be conducted by DLPIU in the 14 earthquake affected districts for all the DLPIU deployed engineers.
- Orientation program on Correction and Exception manual for POs will be conducted by HRRP in the 14 earthquake affected districts.
- HRRP is planning to have a technical workshop focusing on the role of social mobilizers in reconstruction.

Reference Documents

- [Correction exception manual - English version](#)
- [Design Catalogue Volume II- English](#)
- [Updated Inspection Forms](#)
- [Load Bearing Type Design \(NRA/MoUD-CLPIU/JICA/SONA\)](#)
- [MoUD CL-PIU Type Designs: single storey RCC structures \(structure 1, structure 2, structure 3, architecture 1, and architecture 2\), random rubble masonry \(architectural, detail 2, and detail 3\), and dressed stone masonry \(architectural, dressed stone masonry 1, dressed stone masonry 2, and dressed stone masonry 3\).](#)
- [HRRP Weekly Bulletins](#)
- [HRRP FAQs](#)
- [HRRP Briefing Pack](#)

Contact Details

Contact details for the presenters are provided below to facilitate follow up on any of the topics covered in the session.

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For details on the orientation on correction and exception manual for POs please contact the [HRRP team](#) of the respective district.